WORKED SOLUTION

STEP III 1999, Question 12

Probability and Statistics • Discrete Joint Distributions

STEP III 1999, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s3-q12

The key idea

The joint probability depends on j and k mostly through $j+k$, so group the terms by the total $n = j+k$. What is left is $\sum_j \frac{1}{j!(n-j)!}$, which is a binomial sum in disguise: multiply by $n!$ and it becomes $\sum_j \binom{n}{j} = 2^n$. After that, every part is an exponential series.

Solution

Part (i)

Let $n \geq 1$. The event $\{X+Y=n\}$ is the disjoint union of the events $\{X=j,\ Y=n-j\}$ for $j = 0, 1, \ldots, n$. On each of these $j+k=n$, so

$$\begin{aligned} \mathrm{P}(X+Y=n) &= \sum_{j=0}^{n} \mathrm{e}^{-1} \frac{n\lambda^n}{j!\,(n-j)!} = \mathrm{e}^{-1}\frac{n\lambda^n}{n!}\sum_{j=0}^{n}\binom{n}{j} \\ &= \mathrm{e}^{-1}\frac{n\lambda^n 2^n}{n!} = \frac{\mathrm{e}^{-1}(2\lambda)^n}{(n-1)!}, \end{aligned}$$

using $\sum_j \binom{n}{j} = (1+1)^n = 2^n$ by the binomial theorem, and $n/n! = 1/(n-1)!$ for $n \geq 1$. For completeness, $\mathrm{P}(X+Y=0) = \mathrm{P}(X=0,\ Y=0) = 0$, because of the factor $j+k$.

Part (ii)

The probabilities $\mathrm{P}(X+Y=n)$ over all $n \geq 0$ must add up to 1. The $n=0$ term is zero, so

$$1 = \sum_{n=1}^{\infty} \frac{\mathrm{e}^{-1}(2\lambda)^n}{(n-1)!} = \mathrm{e}^{-1}(2\lambda)\sum_{m=0}^{\infty}\frac{(2\lambda)^m}{m!} = \mathrm{e}^{-1}\,2\lambda\,\mathrm{e}^{2\lambda},$$

where we put $m = n-1$ and used the exponential series $\sum_m t^m/m! = \mathrm{e}^t$. Hence $2\lambda\,\mathrm{e}^{2\lambda-1} = 1$, as required.

Part (iii)

Let $g(x) = 2x\,\mathrm{e}^{2x-1}$. By the product rule,

$$g'(x) = 2\mathrm{e}^{2x-1} + 4x\,\mathrm{e}^{2x-1} = 2(1+2x)\,\mathrm{e}^{2x-1}.$$

For $x > 0$ both $1+2x$ and e^{2x-1} are positive, so $g'(x) > 0$ and g is strictly increasing on $x > 0$.

> **Ask yourself** Why does "increasing" give "at most one solution"?

If $0 < x_1 < x_2$ then $g(x_1) < g(x_2)$, so g cannot take the value 1 at two different points. Since $\lambda > 0$, the equation $g(\lambda) = 1$ from (ii) therefore has at most one solution.

Now try the obvious candidate that makes the exponent zero, $x = \frac{1}{2}$: $g(\frac{1}{2}) = 1 \cdot \mathrm{e}^0 = 1$. So $\frac{1}{2}$ is a solution, and by uniqueness it is the only one:

$$\lambda = \tfrac{1}{2}.$$

With this value, part (i) becomes $\mathrm{P}(X+Y=n) = \dfrac{\mathrm{e}^{-1}}{(n-1)!}$ for $n \geq 1$.

Part (iv)

Since $X+Y$ takes only the values $n \geq 1$, with the probabilities just found,

$$\mathrm{E}(2^{X+Y}) = \sum_{n=1}^{\infty} 2^n \frac{\mathrm{e}^{-1}}{(n-1)!} = 2\mathrm{e}^{-1} \sum_{m=0}^{\infty} \frac{2^m}{m!} = 2\mathrm{e}^{-1}\mathrm{e}^2 = 2\mathrm{e},$$

again with $m = n-1$ and the exponential series (which converges, so the expectation is finite).

Checks

- With $\lambda = \frac{1}{2}$, the formula in (i) says $X+Y-1$ has a Poisson distribution with mean 1. Its probabilities $\mathrm{e}^{-1}/m!$ certainly sum to 1, consistent with (ii).
- Summing the joint probabilities numerically over $j+k < 60$ gives total 1.000000, and summing $2^{j+k}\,\mathrm{P}(X=j, Y=k)$ gives 5.436564, which matches $2\mathrm{e} \approx 5.436564$.
- A quick sense check on (iv): $2^{X+Y} = 2 \cdot 2^N$ with $N \sim \mathrm{Po}(1)$, and $\mathrm{E}(s^N) = \mathrm{e}^{s-1}$ for a Poisson variable, so the answer is $2\mathrm{e}^1$.

Common slips

- Forgetting that $n/n! = 1/(n-1)!$, which leaves an awkward factor of n inside the series in (ii).

- Treating the $n = 0$ term as nonzero: the factor $j + k$ makes it vanish, which is exactly why the sums start at $n = 1$.
- In (iii), showing only that $\lambda = \frac{1}{2}$ works. You also need monotonicity to rule out any other solution.

Beyond STEP

The quantity $\mathrm{E}(s^{X+Y})$ is the *probability generating function* of $X + Y$, here evaluated at $s = 2$. Generating functions turn sums of probabilities into power series, and they are the standard tool in a first university probability course for identifying distributions and finding moments. Grimmett and Welsh, *Probability: An Introduction*, gives a clear account of generating functions and the Poisson distribution.

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